



Managing deepfakes with artificial intelligence: Introducing the business privacy calculus

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ABSTRACT

This paper explores the profound implications of artificial intelligence-driven deepfake technology. We introduce a novel business privacy calculus model by delving into the impact of deepfakes through a qualitative explanatory study involving twenty-seven bank managers from three global banks across nine countries. Building on psychological reactance and privacy calculus theories, the evidence shows how data integrity can mitigate deepfake threats, manage business risks, and ensure operational continuity. We propose an AI system architecture that operationalizes responsible AI practices aligned with the business privacy calculus framework. The study contributes to understanding deepfake threats and facilitates the development of a privacy-centric framework for AI governance to safeguard businesses, consumers, and all stakeholders more widely.

1. Introduction

Imagine receiving a letter of legal action from a bank for defaulting on a loan you never took. In another instance, you were denied entry to a country or issuance of a visa due to reports of using a forged passport. A cybercriminal utilized deepfake technology to create a passport with your details, enabling another person to use your identity and reside in that country. Human trafficking, money laundering, blackmail, revenge porn, and ransomware are some of the threats with deepfakes that anyone can fall victim to. This is the new normal of deepfakes powered by artificial intelligence (AI). AI has significantly advanced, revolutionizing business operations (De Bock et al., 2023; Filieri, et al., 2022). As technology evolves and entry barriers decrease, the proliferation of deepfakes will accelerate, driven by the lower creation cost and universal availability (Stroebel et al., 2023).

AI-generated deepfakes threaten people's social lives and businesses by cloning real-life situations undetectably, even by experts (Das et al., 2024; Gambín et al., 2024). Deepfake technology has led to a surge in fraudulent activity. For example, cybercriminals deceived a finance professional into paying US\$ 25 million by impersonating the chief financial officer of a multinational firm through a video conference (Chen & Magramo, 2024). Also, in Hong Kong, perpetrators used eight stolen identity cards to fraudulently make 90 loan applications and 54 bank account registrations within three months (Chen & Magramo,

2024).

In business, managers are frequently presented with a choice: either pay a fee to halt the dissemination of a deep fake or endure severe public repercussions (Kietzmann et al., 2020). A significant deepfake alteration emerged in June 2019, involving a fabricated video featuring Facebook CEO Mark Zuckerberg. In the video, Zuckerberg purportedly conversed about controlling vast amounts of individual data, with the CBSN logo suggesting an association with the news network (Harris, 2019).

Given the potential financial impact on organizations, exploring regulations, security, and privacy risks and addressing any inadequacies in AI infrastructure is imperative (Ahmadi, 2023; AL-Dosari et al., 2024; Rahman et al., 2023). The EU Artificial Intelligence Act (AIA) is a significant legislation designed to revolutionize AI systems. Its primary goal is to shift AI systems towards a model where humans play a crucial role in decision-making processes (Siegel et al., 2024). This paper explores the necessary actions that can be taken from business perspectives to prevent the risk of deepfakes and capitalize on the opportunities presented by AI systems. A significant research gap exists in examining strategies to ensure the privacy protection of consumers, businesses, and society from AI developments (Flavián et al., 2024; Gambín et al., 2024; Karpinska-Krakowiak & Eisend, 2024; Masood et al., 2023). Also, scholars can explore alternative methods to ease tensions arising from asymmetric relationships due to AI (De Bock et al., 2023; Kaur et al., 2024; Sahoo et al., 2024).

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The rapid advancement of AI technologies, particularly deepfakes, presents new challenges to businesses regarding data integrity, privacy, and operational security (Barnes & de Ruyter 2022; Grewal et al., 2021; Heidari et al., 2024; Passos et al., 2024). Traditional data security frameworks fail to address these emerging threats, highlighting the need for novel models and strategies (Heidari et al., 2024; Karpinska-Krakowiak & Eisend, 2024; Kaur et al., 2024). This study aims to bridge the gap between technological advancements and effective business privacy practices by drawing qualitative insights from industry professionals.

Given the novelty of deepfakes and their significant yet not fully understood implications for business privacy risk management (Khan et al., 2024; Li & Wan, 2023; Sharma & Tiwari, 2024), an exploratory approach is essential. Our research focuses explicitly on the implications of deepfake technology within the banking sector, addressing threats to data integrity and business operations. We propose a new business privacy calculus model that integrates psychological reactance and privacy calculus theories to manage these risks effectively (see Fig. 1). To develop this model, we conducted a qualitative explanatory analysis involving twenty-seven bank managers from three global banks across nine countries. We also investigated how businesses can implement AI-driven solutions aligned with privacy-centric governance frameworks. Employing qualitative methods from Gioia et al. (2013) and Magnani & Gioia (2023), we systematically analyzed interview data from bank managers, grounding our findings in their experiences.

Despite the growing relevance of deepfakes, existing research has predominantly focused on technical and legal aspects, leaving a critical gap in understanding the impact on business (Ahmadi, 2023; Heidari et al., 2024; Masood et al., 2023; Sivathanu et al., 2023). Furthermore, the literature reveals a limited exploration of privacy calculus in organizational data management, primarily focusing on the cost-benefit trade-off without fully addressing emerging threats (Cloarec et al., 2024; Kezer et al., 2022; Meier & Krämer, 2024; Willems et al., 2023). Additionally, while psychological reactance theory highlights the

impact of deepfakes on individuals' autonomy and decision-making freedom, its application to business decision-making is limited (Campbell et al., 2022; Mühlhoff, 2023).

The paper identifies a significant research gap in understanding how organizations, particularly in the banking sector, can effectively manage the risks posed by deepfakes through the lenses of privacy calculus and psychological reactance. This study addresses the call from scholars (Jabbar et al., 2023; Kaur et al., 2024; Masood et al., 2023; Passos et al., 2024; Sharma & Tiwari, 2024) to investigate how deepfakes impact organizational data management and introduce robust frameworks to manage deepfake risk.

Overall, the key contributions of the work are i) the introduction of the novel business privacy calculus model, combining the impact of psychological reactance and privacy calculus theories, ii) the extending the psychological reactance theory from the consumer to managers' perspective manifesting a business context, and iii) recommending the use of AI to mitigate deepfake risk for a business to safeguard broader stakeholders.

2. Theoretical background

2.1. Deepfakes

Deepfake technology, driven by sophisticated deep learning (DL) and computer vision techniques, presents substantial risks to business decision-making processes (Heidari et al., 2024). By generating highly realistic simulations of real-life situations, individuals, and events, deepfakes challenge organizations' abilities to distinguish authentic content from manipulated media (Kaur et al., 2024; Passos et al., 2024). The capacity of deepfake technology to produce hyper-realistic fabricated content, even advanced detection systems, can be deceived (Flavián et al., 2024; Siegel et al., 2024; Sivathanu et al., 2023).

Deepfakes utilize deep learning, a machine learning technique that leverages deep neural networks (DNNs) inspired by biological brains'

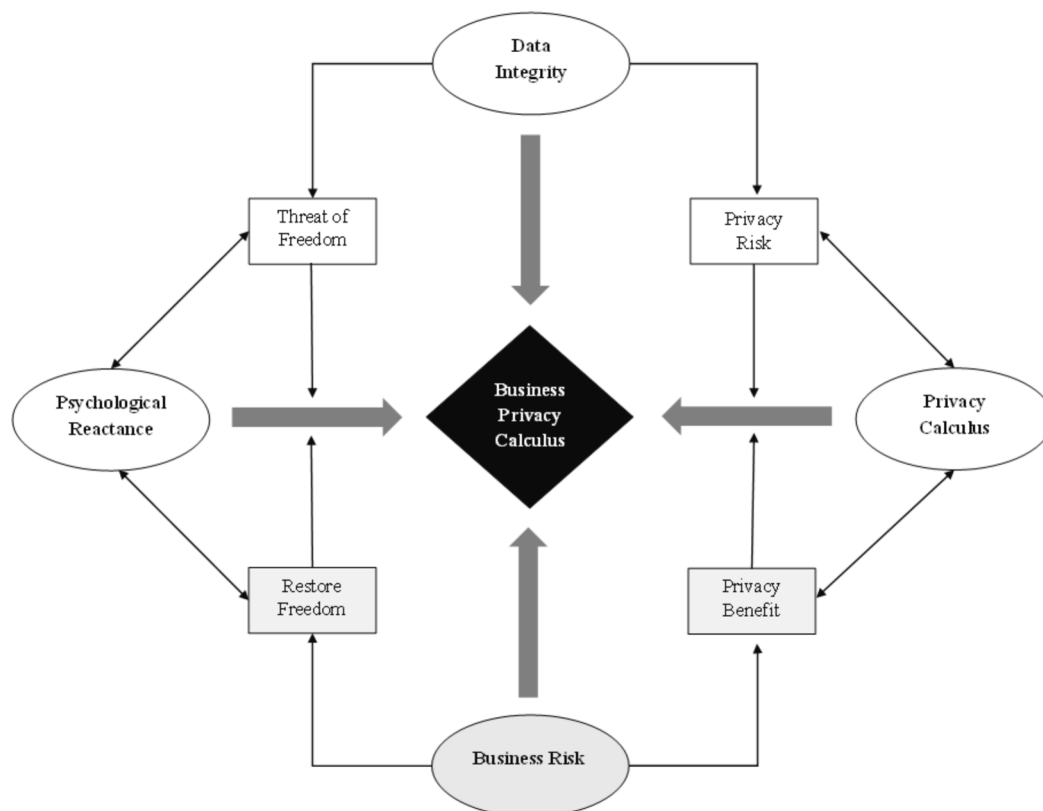


Fig. 1. The proposed business privacy calculus model.

structure and communication processes. These DNNs manipulate images, audio, and video to create hyper-realistic fabricated content (Passos et al., 2024).

The impact of deepfakes spans various industries, eroding trust, security, and privacy (Khan et al., 2024; Stroebel et al., 2023). In the entertainment industry, deepfakes can create damaging fake celebrity videos, leading to reputational harm and misinformation. Deepfakes pose a severe threat in the political realm by manipulating public opinion and endangering democratic processes (Kietzmann et al., 2020). The healthcare sector faces risks from deepfake-generated medical content, which could result in misdiagnoses. In the banking sector, deepfakes introduce unique challenges, including creating fake financial documents and manipulating video calls to impersonate clients (Sharma & Tiwari, 2024). Furthermore, cybercriminals have exploited deepfake technology to create synthetic identity fraud (Siegel et al., 2024).

Despite the growing concern, existing research has largely overlooked the impact of deepfakes on business decision-making processes (Karpinska-Krakiwiak & Eisend, 2024; Sharma & Tiwari, 2024). The literature predominantly addresses consumer perspectives, often neglecting how deepfakes undermine organizational integrity and decision-making frameworks (Dwivedi et al., 2021; Sahoo et al., 2024). Managers increasingly struggle to differentiate between authentic and deepfake-manipulated content, resulting in strategic errors, financial losses, and reputational damage (Gambín et al., 2024; Heidari et al., 2024). The anonymity and ease of creating deepfakes exacerbate these risks, complicating manipulated content's traceability and prevention (Kwok & Koh, 2021; Mukta et al., 2023).

The democratization of deepfake creation tools further complicates detection efforts. As these technologies become more accessible, ensuring content authenticity becomes increasingly challenging. To address the threat of deepfakes, particularly in sensitive sectors such as banking, it is crucial to integrate advanced data science, analytics, and visual recognition technologies into organizational solutions (Masood et al., 2023). Although research has explored technical and legal aspects of deepfakes, the organizational impacts remain under-researched (de Rancourt-Raymond & Smaili, 2023; Kaur et al., 2024; Passos et al., 2024). This gap highlights the urgent need for further investigation into how businesses can adapt to the evolving threat landscape posed by deepfakes, balancing risks and opportunities (Busacca & Monaca, 2023; Karpinska-Krakiwiak & Eisend, 2024).

Organizations must enhance their capabilities to detect and respond to deepfake content to safeguard their decision-making processes from digital deception (Cochran & Napshin, 2021; de Rancourt-Raymond & Smaili, 2023). Future research must develop robust detection and verification methods that ensure data reliability in this increasingly complex threat environment (Heidari et al., 2024; Li & Wan, 2023; Sivathanu et al., 2023). This study focuses on the banking industry to examine the impact of deepfakes on business.

Focusing on the banking sector is particularly critical due to its pivotal role in the global economy. Banking decisions significantly influence other industries, as financing is fundamental to business operations and growth (Bateman, 2020; Khan et al., 2024; Porfirio et al., 2024; Sharma & Tiwari, 2024). The sector's reliance on accurate customer records for credit assessments, identity verification, and fraud detection makes it highly vulnerable to deepfake risks (Al-Dosari et al., 2024). Deepfake directly jeopardizes data accuracy, potentially leading to financial losses, legal liabilities, and reputational damage (Siegel et al., 2024). Moreover, deepfakes complicate the banking industry's stringent compliance requirements, making the authentication of digital records increasingly challenging (Khan et al., 2024).

Investigating the impact of deepfakes on the banking sector allows for a comprehensive understanding of how these risks intersect with regulatory requirements and provides strategies to mitigate these challenges (Chen & Magramo, 2024). Given the central role of banking in the global economy, deepfake-related risks could have far-reaching consequences, potentially destabilizing financial markets and entire

economies (Ebirim & Odonkor, 2024; Sharma & Tiwari, 2024; Bateman, 2020; Porfirio et al., 2024). These risks extend to other industries, such as healthcare, insurance, and government services, all of which are influenced by financial procedures applied by banks (Liyanaarachchi et al., 2021; Porfirio et al., 2024; Shams et al., 2020). Addressing these risks fills a critical void in academic research and offers practical insights for maintaining global economic stability.

2.2. Privacy calculus

Privacy calculus originated from initial works by Laufer and Wolfe (1977) on consumer rational choice based on cost-benefit trade-offs. Extending this trade-off, Culnan and Armstrong (1999) introduced the idea to privacy literature on disclosure decisions. Privacy calculus depicts consumers weighing perceived benefits and risks before disclosing personal information. As technology evolves, particularly AI, the significance of privacy calculus becomes even more pronounced with the emerging risks of deepfake technology (Kezer et al., 2022). Luo et al. (2023) emphasize the need to examine how individuals evaluate factors such as the trustworthiness of AI systems, the potential consequences of data misuse, and the impact on privacy calculus. Consequently, it has become imperative for firms to understand consumers' acceptance of information collection (Hardey & Atkinson, 2018).

We argue that organizations neglecting privacy issues in this era of artificial intelligence can significantly impact business value. Consumers face increased concerns about privacy as AI algorithms generate convincing fake content (Spiekermann et al., 2022). In this context, the privacy calculus framework offers valuable insights about risk-benefit trade-offs in AI-driven environments (Willems et al., 2023). A primary impact of deepfakes on privacy calculus theory is the erosion of trust and confidence in digital transactions (Jabbar et al., 2023). Also, advancements in AI systems and the exponential growth of data collection pose new challenges to the privacy calculus (Luo et al., 2023).

A notable literature gap exists due to the narrow focus on privacy calculus centered on the cost-benefit trade-off (Beke et al., 2022; Meier & Krämer, 2024). Hence, examining the concept in organizational data management is useful (Kezer et al., 2022; Willems et al., 2023). Future research can explore the relationship between firm-customer relationship dynamics on privacy calculus over time (Jabbar et al., 2023; Wang et al., 2024). This research aims to fill the gap by examining the privacy calculus from the perspective of organizations on data management processes with deepfake risk.

2.3. Psychological reactance theory

Psychological reactance theory illustrates how individuals respond to perceived threats invading their autonomy and freedom (Brehm, 1966). When faced with a threat to freedom, people encounter a state of reactance, compelling them to resist actively reclaiming freedom. They can also respond with anger and counteraction, restoring freedom (Badewi et al., 2023). The emergence of deepfake technology poses significant threats to individuals' autonomy and freedom in decision-making processes, eliciting psychological reactance (Campbell et al., 2022). Hence, the proliferation of deepfake videos and AI-generated content exacerbates concerns about misinformation and manipulation, fostering psychological reactance among users (Lim et al., 2023; Wong et al., 2024; Xie et al., 2024).

These concerns extend to institutions where deepfakes jeopardize the accuracy of risk assessment processes by manipulating financial data and executive communications (Masood et al., 2023). With AI influence, the freedom to make decisions is threatened due to informational power asymmetry between a few dominant players in business and society (Mühlhoff, 2023). In the context of institutions, the proliferation of deepfake videos and AI-generated content exacerbates concerns about deception, fostering reluctance to use systems (Masood et al., 2023). Managers may respond to these threats by overcompensating with

subjective judgment over data-driven insights (de Rancourt-Raymond & Smaili, 2023).

Additionally, threats to data integrity and trust in customer relations further compromise managers' freedom to maintain transparent and trustworthy social interactions (Giantini, 2023). Restoring freedom through investment in advanced technologies for detecting and preventing Deepfake is critical to regain control over decision-making processes.

Concerns about digital privacy heighten psychological reactance, potentially compromising decision-making and autonomy in managing data (Liyanaarachchi et al., 2024). Organizations must balance security measures and privacy concerns, deploying AI technologies transparently and ethically (Sontan & Samuel, 2024). This research explores deepfakes' impact on business decision-making, focusing on triggered psychological reactance.

3. Methodology

3.1. Research approach

Our qualitative study addresses a critical gap in privacy calculus research by introducing the "business privacy calculus" framework. While existing theories predominantly focus on individual perspectives, our research shifts to the organizational level, investigating how businesses evaluate privacy risks and benefits, especially regarding emerging threats like deepfakes. This shift aligns with calls for a broader understanding of privacy calculus beyond individual concerns (Beke et al., 2022; Cloarec et al., 2024; Sah & Jun 2024). Furthermore, we extend psychological reactance theory by shifting from the current focus on individual perspectives to exploring how increased privacy risks from deepfakes heighten reactance among managers.

This shift highlights how such risks impact managerial decision-making autonomy (Badewi et al., 2023; Ogbanufe & Gerhart, 2022). Utilizing the qualitative methods of Gioia et al. (2013) and Magnani & Gioia (2023), we systematically analyzed interview data from bank managers, ensuring that our findings are grounded in participants' experiences. This approach enabled us to develop a new theory demonstrating how our novel insights contribute to a revised understanding of how organizations manage privacy risks and respond to emerging threats.

We initially approached five major global banks to participate in our study on deepfake risks in banking. After comprehensive correspondence and evaluation, we selected three banks that best aligned with our research objectives and demonstrated substantial expertise in addressing these risks. This meticulous selection process ensured that our study included institutions with the most relevant experience and a broad international presence, thereby enhancing the validity and impact of our findings.

We conducted in-depth interviews with twenty-seven bank managers from these three banks, each with at least ten years of experience. The managers were deliberately chosen from nine countries: the United States, United Kingdom, Sri Lanka, Hong Kong, Australia, UAE, Canada, Malaysia, and India. Anchored in the codification of participants' interviews (Gioia et al., 2013; Magnani & Gioia, 2023), our sampling strategy utilized purposive selection to ensure that participants had direct experience with deepfake risks, as Holland (2005) outlined. This approach was meticulous and purposeful, consistent with the methodology described by Campbell et al. (2020). By engaging with the selected banks' Cybersecurity and Fraud Prevention divisions, we covered a diverse range of financial markets, regulatory environments, and technological maturity levels.

Additionally, selecting emerging markets like India, Malaysia, and Sri Lanka provided valuable insights into how rapidly growing digital economies adapt to these risks. This strategic selection offered a comprehensive view of advanced and adaptive deepfake prevention approaches.

Given our sample's diverse international backgrounds, we determined that a larger sample size of twenty-five participants was necessary to achieve code saturation, compared to the typical range of 9–17 participants in similar studies (Hennink & Kaiser, 2022). We also conducted two extra interviews to ensure member validation, enhancing our findings' accuracy and credibility, as Bell et al. (2022) recommended. This rigorous approach aligns with the standards set by Gioia et al. (2013) for ensuring the robustness of qualitative research.

Interviews were conducted online from September 2023 to March 2024, with durations ranging from 47 to 82 min each. To ensure convenience and accessibility, interviews were conducted via Zoom videoconferencing, guaranteeing interviewee anonymity to promote candid responses and open discussions. Due to bank policies and our agreement with the participating institutions, the interviews were not recorded. Instead, four researchers conducted each interview and took detailed notes. These notes were subsequently compiled and shared with the research team to ensure clarity and a common understanding of the discussions (Magnani & Gioia, 2023). This collaborative approach facilitated accurate data analysis and coding, maintaining the integrity and consistency of the research process.

The key research questions were: 1) Can you describe your concerns about the Deepfake risks in using customer data?, ii) Can you share your experiences with any training on deepfakes or the privacy aspects of AI-related systems? How has this training (or lack thereof) affected your understanding or approach to these issues? iii) Can you describe the processes in your bank regarding deepfake detection? Who typically takes responsibility, and how is this role defined or supported within the organization?, iv) Can you elaborate on the protocols, controls, and standards your organization has implemented to detect and prevent deepfakes? How do these measures work in practice, and what challenges have you encountered, v). In what ways have deepfakes influenced or changed job roles within your organization? Can you share specific examples of how responsibilities or expectations have shifted?, vi). How do you perceive the significance of addressing and identifying deepfakes within your organization? Can you discuss whether this issue has become a business priority and why?.

3.1.1. Data analysis

The data analysis utilized a three-tiered coding system following the Gioia method (Gioia et al., 2013). The Gioia method was chosen for its efficacy in uncovering nuanced insights and facilitating theory development from empirical data. Three coders manually validated the coding, assessing agreement levels using a percentage agreement method. The intercoder agreement revealed an 89 % overlap between coders. Subsequently, the remaining 11 % of data was deliberated among the coders until a consensus was reached. Member checks were performed with eight participants to enhance internal validity and validate thematic accuracy (Guba & Lincoln, 1994).

During the first stage, interviews were coded and organized into a series of first-order concepts from managers' and business perspectives. For managers, perspective is indicated by concepts such as resistance to decision-making, reactions to freedom threats, autonomy, and restoring decision-making. From a business perspective, codes included the data value, authenticity, heightened data privacy risks, uncertain privacy benefits, lack of validation, business returns, transparency, and the trade-off between privacy risk and return. The first-order concepts were further analyzed to derive open codes, leading to 2nd order themes building on theoretical grounding (Corbin & Strauss, 2014).

Subsequently, a second-order analysis was conducted to establish connections between privacy calculus and psychological reactance, leading to new concepts of data integrity and business risk. Further, each interview was analyzed using a constant comparison method, connecting to the emergence of the business privacy calculus model based on privacy calculus and psychological reactance theories (Braun & Clarke, 2021). In the final stage, key themes were synthesized into a newly formulated conceptual model, "The business privacy calculus." This

approach facilitated exploring participants' experiences and business contexts through the lens of research questions embedded within the coding system (Gioia et al., 2013). Tables 1 and 2 present the data structure and the selective quotes of respondents.

3.1.2. Findings

3.1.2.1. Data integrity. Data integrity is pivotal for data accuracy, consistency, and reliability for business decisions (Ahmed et al., 2023; Cozzolino et al., 2022). The emergence of deepfake technology poses a formidable challenge to data integrity (Abbas et al., 2023). Given the evolving threat landscape, bank managers must adopt proactive measures to safeguard data integrity through advanced AI systems (Hossain et al., 2024; Shams et al., 2020).

Two sub-categories, a threat to freedom and privacy risk, emerged under the selective code data integrity. Managers express concerns about the impact of compromised data integrity on their decision-making freedom. *"I fear my decisions can encourage deepfake action. Any decision based on deepfake will result in fraud or cybercrime, as it will benefit a hacker at the expense of an innocent customer"* (B2, 4). The higher privacy risk associated with compromised data integrity due to malicious actors' potential for manipulation and misuse outweighs any

perceived benefits. *"The idea of deepfakes makes me feel uneasy. I have to handle customer requests using our systems and respond quickly. I do not have the luxury of spending much time verifying each request. So, I cannot keep the privacy risk under control because it is much higher than what we find acceptable"* (B3, 2).

3.1.2.1.1. Threat of freedom. Managers' sentiments echo the essence of psychological reactance where the threat of freedom becomes evident as deepfakes compromise data integrity. *"With deepfakes, I am unsure about the validity of the data I use. This has diminished my sense of freedom in my work environment"* (B2,6). This uncertainty reflects the erosion of autonomy and the pervasive concern that deceptive information may influence or distort decision-making processes. *"I find myself questioning the validity of the data. This seriously impacts my autonomy in decision-making. I feel hampered by the uncertainty surrounding the authenticity of information. I am not free in thinking like I used to be"* (B1,1).

Furthermore, managers expressed that the threat of deepfakes obstructs autonomous thinking and decision-making processes. *"The presence of deepfakes makes me uneasy. I am constantly second-guessing my decisions, fearing I might make a blunder. That is the end of my career. I have never felt this uncomfortable in the last 12 years of my career"* (B3,4).

"The threat of deepfakes undermines our autonomy. We are powerless and vulnerable to external manipulation and the fear of being controlled by

Table 1
Data structure.

1st order concepts	open codes	2nd order themes	Theoretical grounding	Aggregate dimensions	Iterative process for theory building	Substantive theory
Privacy risk is at an uncontrollable level. Data integrity is impossible with higher privacy. Higher threat of privacy invasion with deepfakes. Systems are unable to detect deepfakes. Privacy protection and resources are inadequate. Higher risk of AI-driven cybercriminal activity.	Data breach Privacy invasion Integrity	Privacy risk	Psychological Reactance Privacy Calculus	Data integrity	Consider consumer data as a resource. Use privacy calculus as part of the decision-making process. Manage psychological Reactance for privacy protection. Cultivate a culture of data stewardship. Recognize correlation between data integrity and business risk. Inbuilt system for deepfake detection	Business Privacy Calculus
Deepfake risk diminishes freedom at work. Deepfakes create uncertainty about data integrity. Sense of unease and caution due to deepfakes. Fear of succumbing to manipulation and deception. Confinement in the decision-making process. distinguishing deepfakes is impossible.	Freedom Threat Integrity Ambiguity	Threat of freedom			Develop AI to counter deepfake risks. Design a business model centred on privacy Training to detect deepfake threats	
Decision-making protected with liability coverage. Deepfake mitigation to enhance autonomy. Strong internal communication to foster resilience. Training in AI adaptability. Data assurance through management support. Automatic deepfake detection.	Autonomy Liability cover Cyber intelligence Conviction	Restoration of freedom		Business risk		
Inbuilt deepfake detection system for data integrity. Ensure data resilience, prioritizing data privacy. Strengthening the data authentication processes. Enhance data value to mitigate privacy risks. Data assurance to manage inherent deepfakes risk. AI-driven real-time customer verification.	Dara resilience Data valuation System intelligence Validation	Privacy benefit				

Table 2
Selective quotes and 2nd order themes.

2nd order themes	Selective quotes from respondents				
Privacy risk	<i>The rising sophistication of deepfakes is seriously worrying. It is increasing our privacy risk because we are not sure if the data can be trusted anymore. The risk is beyond our tolerance limit"</i>	"Privacy risks are at an all-time high with the proliferation of deepfakes, especially when protecting our customers' data integrity"	<i>"Escalating privacy risks by deepfakes leaves us in a precarious position. Without assurances of customer data authenticity, we are navigating uncharted waters where benefits seem increasingly elusive"</i>	"As a manager, increasing privacy risks with deepfakes is a serious issue. With uncertainty surrounding data legitimacy, the risks far outweigh any potential rewards"	<i>"In banking, risk management is like walking on a tightrope. Authentic customer data assurance is vital; without it, we're uncertain as we are accountable, especially for our lending decisions. Deepfakes risk is beyond our control"</i>
Threat of freedom	"With deepfakes, I am unsure about the validity of the data I use. This has diminished my sense of freedom in my work environment"	<i>"I find myself questioning the validity of the data. This seriously impacts my autonomy in decision-making. I feel hampered by the uncertainty surrounding the authenticity of information. I am not free in thinking like I used to be"</i>	"The presence of deepfakes makes me uneasy. I am constantly second-guessing my decisions, fearing I might make a blunder. That is the end of my career. I have never felt this uncomfortable in the last 12 years of my career"	"The threat of deepfakes undermines our autonomy. We are powerless and vulnerable to external manipulation and the fear of being controlled by outsiders"	<i>"Just the thought of deepfakes freaks me out, mate. I am lost, honestly, and worried that I might be trapped with a dodgy character. You never know. I can't trust any data these days. It feels like the future of banking's looking pretty grim"</i>
Restoration of freedom	<i>"We should implement built-in systems with advanced detection capabilities and provide real-time alerts and insights to help us navigate deceptive content"</i>	"We urgently need strategies to restore autonomy in decision-making without fear of data integrity threats. Also, there should be advanced training on these AI-based privacy issues to support our decisions"	<i>"Mate, reckon if we have job security for any mistakes influenced by those dodgy deepfakes, it will give us a fair boost. We would feel heaps more secure in our choices and have the freedom to crack on"</i>	<i>"Proactive measures, like liability insurance on data-driven decisions, could ensure confidence in my ability to do my job properly"</i>	"As a manager, I must exude positivity, clarity, and confidence in decision-making, especially amid deepfake threats. Our ability to make sound judgments relies on freedom from deepfake fears"
Privacy benefit	<i>."Investing in AI-driven real-time customer verification helps us enhance data value while addressing deepfake privacy risks. The bank should have a higher return on investment regarding the data value than the privacy cost"</i>	"We must control customer data to maximize its potential while ensuring data cleanliness and transparency to avert financial losses and business disruptions caused by fake data"	."We are accustomed to business risks like bad loans and customer financial concerns, but deepfake risk extends beyond normal parameters. Addressing this risk is crucial for the industry's survival"	<i>"An interactive strategy involving training, real-time system upgrades, and collaboration with international cybercrime agencies to mitigate deepfake risks is vital"</i>	"Getting on board with AI-driven customer verification beefs up our data's worth and keeps deepfakes under control. Also, an additional safety net to knock out risks before they even get a chance to kill the whole system."

outsiders“ (B2, 3). The fear of deceptive information erodes confidence in decision-making and autonomy, highlighting the necessity for proactive measures to safeguard data integrity in business operations. *“Just the thought of deepfakes freaks me out, mate. I am lost, honestly, and worried that I might be trapped with a dodgy character. You never know. I can't trust any data these days. It feels like the future of banking's looking pretty grim”* (B1,7).

3.1.2.1.2. *Privacy risk.* Managers experience higher privacy risks associated with compromised data integrity, as data misuse outweighs perceived benefits. The participants' sense of unease about the increasing sophistication of deepfakes resonates with privacy calculus theory: *“The rising sophistication of deepfakes is seriously worrying. It is increasing our privacy risk because we are not sure if the data can be trusted anymore. The risk is beyond our tolerance limit”* (B3,5).

This highlights the constant vigilance required to prevent privacy breaches and underscores the need for proactive measures to mitigate risks. *“Privacy risks are at an all-time high with the proliferation of deepfakes, especially when protecting our customers' data integrity”* (B1,9). Without assurances of customer data authenticity, the potential benefits of banking operations are overshadowed by the heightened data privacy risk. *“Escalating privacy risks by deepfakes leaves us in a precarious position. Without assurances of customer data authenticity, we are navigating uncharted waters where benefits seem increasingly elusive”* (B3,8). *“As a manager, increasing privacy risks with deepfakes is a serious issue. With uncertainty surrounding data legitimacy, the risks far outweigh any potential rewards”* (B2,9). *“In banking, risk management is like walking on a tightrope. Authentic customer data assurance is vital; without it, we're uncertain as we are accountable, especially for our lending decisions. Deepfakes risk is beyond our control”* (B3,1).

3.1.2.2. *Business risk.* The proliferation of deepfakes poses significant business risks and threatens business continuity (Flavián et al., 2024),

and it can incite illegal activities (Busacca & Monaca, 2023; Liyanaarachchi et al., 2023). Deepfakes can manipulate customer data for credit assessments or identity verification, leading to inaccurate risk assessments and financial losses (AL-Dosari et al., 2024; de Rancourt-Raymond & Smaili, 2023). Moreover, fake financial analysis or false statements about business performance can lead to economic losses for investors (Di Domenico & Visentin, 2020; Rahman et al., 2023).

The findings indicate that a lack of data integrity increases business risk (Rahman et al., 2023). Similarly, two sub-categories, restoration of freedom and privacy benefit, emerged under the selective code business risk. *“As a manager, I have to be positive, clear, and confident in my decision-making, especially in the face of deepfake threats. My ability to make sound judgments hinges upon our freedom from the fear of deepfake consequences”* (B1,4). *“Deepfakes erode data integrity and the potential for flawed decisions. These jeopardize our financial stability and reputation, amplifying business risks and threatening continuity”* (B2,6). Organizations require systems to safeguard data integrity and mitigate risks across financial, reputational, regulatory, and operational domains in the future.

3.1.2.2.1. *Restoration of freedom.* The higher business risk resulting from inaccurate decisions influenced by deepfakes triggers a reactance to make decisions. To address deepfake threats, it is crucial to support managers in reclaiming their decision-making autonomy and mitigating fears of job loss or financial risks from erroneous choices. This aligns with psychological reactance theory, which suggests defensive reactance to control uncertainty (Xie et al., 2024). Managers can regain autonomy by implementing advanced detection capabilities and offering proactive assurance (Li & Shen, 2022; Sontan & Samuel, 2024).

To address this challenge, it becomes imperative to implement advanced detection capabilities within an organization. *“We should implement built-in systems with advanced detection capabilities and provide real-time alerts and insights to help us navigate deceptive content”* (B2,2).

The managers must be equipped with resources, training, and technology on deepfake detection to empower them to make informed decisions. *“We urgently need strategies to restore autonomy in decision-making without fear of data integrity threats. Also, there should be advanced training on these AI-based privacy issues to support our decisions”* (B1,8).

The compromise of decision integrity by evolving deepfake threats erodes confidence and assurance and undermines certainty in decision-making processes. *“Mate, reckon if we have job security for any mistakes influenced by those dodgy deepfakes, it will give us a fair boost. We would feel heaps more secure in our choices and have the freedom to crack on”* (B3, 3). *“Proactive measures, like liability insurance on data-driven decisions, could ensure confidence in my ability to do my job properly”* (B3,1). This assurance allows managers to focus on their professional expertise rather than forensic data security, enhancing their decision-making effectiveness. *“As a manager, I must exude positivity, clarity, and confidence in decision-making, especially amid deepfake threats. Our ability to make sound judgments relies on freedom from deepfake fears”* (B2,5).

3.1.2.2.2. Privacy benefit. Data is a strategic resource for a business. Therefore, the cost of using data should be less than the benefit derived from such decisions (Willems et al., 2023). Nevertheless, deepfake risks present a formidable challenge, threatening data integrity and diminishing decision-making quality (Luo et al., 2023). Managers recognize the need to effectively manage deepfake risks to maximize the value derived from data usage. This involves implementing robust measures to detect, prevent, and mitigate the impact of deepfakes to manifest business continuity (Busacca & Monaca, 2023). *“Investing in AI-driven real-time customer verification helps us enhance data value while addressing deepfake privacy risks. The bank should have a higher return on investment regarding the data value than the privacy cost”* (B2,5). This mitigates the business risk from compromised data integrity and ensures privacy. Managers stress the significance of controlling customer data to maximize its potential while minimizing the risk of malicious exploitation. *“We must control customer data to maximize its potential while ensuring data cleanliness and transparency to avert financial losses and business disruptions caused by fake data”* (B3,6).

In addition to managing deepfake risks, banks must effectively handle privacy risks to maintain control and clarity amid potential threats. *“We are accustomed to business risks like bad loans and customer financial concerns, but deepfake risk extends beyond normal parameters. Addressing this risk is crucial for the industry’s survival”* (B1,2). *“An interactive strategy involving training, real-time system upgrades, and collaboration with international cybercrime agencies to mitigate deepfake risks is vital”* (B1,5). By taking these steps, an organization can uphold the integrity of its operations, ensuring that the benefits of privacy outweigh the associated risks. *“Getting on board with AI-driven customer verification beefs up our data’s worth and keeps deepfakes under control. Also, an additional safety net to knock out risks before they even get a chance to kill the whole system.”* (B1,4).

3.2. Business privacy calculus

The research process has led to the development of a novel business model that addresses the challenges posed by deepfakes. We introduce the “business privacy calculus,” a strategic framework to manage psychological reactance, generate higher privacy benefits over cost, ensure data integrity, and minimize business risk. By ensuring that the benefits obtained from using customer data outweigh the associated privacy risks, the model enables organizations to navigate the intricate landscape of data privacy. The model transcends the privacy calculus theory from consumer-centric to business-centric perspectives.

We define business privacy calculus as:

The business privacy calculus is a privacy risk–benefit trade-off that prioritizes managerial data integrity and mitigates business risks to ensure uninterrupted business operations.

This framework fosters a culture of responsible data stewardship, where businesses prioritize the ethical and transparent use of customer

data while remaining adaptable to evolving regulatory frameworks and consumer expectations. Adopting this framework ensures regulatory compliance and cultivates an environment of trust and confidence among customers, thereby safeguarding the integrity of business operations in the ever-changing landscape of digital deception.

4. Discussion

4.1. Theoretical implications

This study offers two significant theoretical contributions. First, we introduce the concept of “business privacy calculus,” outlining a unique privacy scenario from an organizational standpoint.

The current focus of privacy calculus is on the individual perspective, often overlooking its impact on businesses (Cloarec et al., 2024; Meier & Krämer, 2024). Our study addresses this gap by introducing the concept of “business privacy calculus,” which posits that organizations, like consumers, use a privacy calculus when handling consumer data, balancing specific privacy risks and benefits. This is particularly critical in deepfakes, where uncertain data authenticity amplifies business risks. Consequently, we argue that the privacy calculus is more complex and heightened for organizations, offering new insights into the existing literature.

There is a call for a broader understanding of privacy calculus beyond the direct risks and benefits perceived by consumers (Beke et al., 2022; Cloarec et al., 2024; Dienlin & Metzger, 2016; Sah & Jun 2024; Wang et al., 2024). However, there remains a significant gap in exploring this from an organizational perspective, especially in light of emerging technologies like deepfakes. Privacy calculus has typically emphasized the individual’s transactional relationship between risks and benefits, but technological advancements have made these interactions far more complex (Meier & Krämer, 2024). As consumer data collection becomes more pervasive, this necessitates a shift in focus to the psychological conflicts between privacy concerns and trust in service providers (Sah & Jun 2024).

Our study’s primary contribution is introducing the “business privacy calculus” framework, which extends the existing literature by providing a comprehensive perspective on privacy risks and benefits from an organizational standpoint. This paradigm shift addresses the impact of deepfakes and offers a strategic model for business decision-making, responding to the scholarly call for a more comprehensive approach to privacy calculus (Beke et al., 2022; Meier & Krämer, 2024). Further offers strategies to counter the negative impacts of technological transformation (Das et al., 2024; Flavián et al., 2024), particularly in the context of AI advancements, an area still under-researched (Mustak et al., 2023; Siegel et al., 2024).

Second, the study recognizes the profound impact of deepfakes on psychological reactance theory, particularly concerning organizational capability in managing data. While existing literature mainly examines reactance from a consumer perspective (Badewi et al., 2023; Ogbanufe & Gerhart, 2022), this research introduces a paradigm-shifting viewpoint. We assert that the heightened privacy risks associated with deepfakes contribute to increased reactance among managers, thereby threatening their freedom in decision-making. This innovative perspective expands reactance theory from the consumer domain to the business sphere, emphasizing that organizations must devise strategies that restore freedom to ensure consistent managerial decisions. The conceptual model proposed in this study provides a strong basis for investigating the impact of advanced technology usage on psychological reactance. (Badewi et al., 2023; Chang et al., 2023).

4.2. Managerial implications

We provide five implications for practice based on the business privacy calculus model. First, organizations should focus on strategic integration for enterprise risk management. To effectively address

deepfake risks, businesses must embed deepfake risk management into their broader risk frameworks and align it with organizational policies and objectives. Utilizing automated content analysis tools, supported by dedicated teams, allows for real-time monitoring of media content, ensuring that decisions are informed by a careful trade-off between risk and benefit, as outlined in the business privacy calculus. This approach ensures that deepfake risks are managed alongside operational risks, enhancing overall organizational resilience (Rahman et al., 2023).

Second, organizations must enhance cybersecurity measures through advanced technological solutions. The business privacy calculus emphasizes the importance of protecting the integrity of business data. Integrating automated detection tools with existing cybersecurity infrastructure and employing cryptographic techniques to verify media authenticity aligns with the model's goal of minimizing risks while ensuring privacy costs do not outweigh business benefits. Regular updates to Security Information and Event Management (SIEM) systems and the implementation of stringent access controls are vital for maintaining the trust and security of digital assets (Gambín et al., 2024).

Third, empowering staff through targeted training is critical for deepfake risk mitigation. According to the business privacy calculus, informed decision-making and minimizing psychological reactance are crucial to maintaining business operations. Comprehensive training programs, including interactive e-learning modules and simulations, equip employees with the knowledge to recognize and respond to deepfake threats effectively. Regular updates to training content ensure that staff are aware of emerging risks, fostering a culture of vigilance and responsible data stewardship (Ahmadi, 2023).

Fourth, proactive crisis response planning is essential for managing deepfake incidents. The business privacy calculus framework underscores the importance of preserving data integrity and organizational reputation. Establishing robust crisis protocols, training teams on advanced forensic techniques, and using deep learning models to differentiate between authentic and manipulated media align with the model's focus on minimizing reputational and financial risks. Secure data storage using Advanced Encryption Standards (AES) further safeguards against potential breaches, ensuring swift and effective responses to deepfake incidents (Busacca & Monaca, 2023; Siegel et al., 2024).

Finally, organizations should actively engage in collaborative industry efforts to combat deepfake risks. The business privacy calculus promotes the idea of collective action to enhance overall resilience. Participating in industry forums, sharing best practices, and working closely with regulatory bodies ensures that organizations stay informed of the latest standards and advancements. This collaborative approach strengthens defenses against deepfakes and aligns with the calculus's objective of fostering trust and maintaining the ethical use of customer data across the digital ecosystem (Lim et al., 2023).

4.3. Limitations and future research avenues

Future research can extend the examination of deepfake threats beyond the banking sector to encompass diverse industries, validating the proposed model and offering insights into industry-specific vulnerabilities and mitigation strategies (De Bock et al., 2023; Stroebel et al., 2023). In the entertainment industry, deepfake technology has been exploited to create fake celebrity pornography, violating privacy and raising consent concerns (Meskys et al., 2020). Similarly, in politics, misleading videos of politicians engaging in unethical behavior can influence public opinion and undermine trust in political processes (Kietzmann et al., 2020).

The business privacy calculus elucidates the intricate interplay between data integrity and business risk in managing deepfake risk in a business context. Future studies could quantitatively generalize and validate the model across broader industry sectors and business environments (De Bock et al., 2023; Stroebel et al., 2023).

Additionally, exploring consumer responses to deepfake risks amid AI advancements and immersive platforms warrants empirical

investigations (Flavián et al., 2024; Gambín et al., 2024; Masood et al., 2023). Scholars can examine the deepfake risk from the perspectives of cybersecurity specialists involved in managing AI applications and including a cross-section of management levels to obtain a holistic understanding (Bray et al., 2023; Camilleri, 2023; Whittaker et al., 2023). Additionally, the model can be tested quantitatively to ensure generalization to manifest AI-driven deepfakes and resulting implications (Bray et al., 2023; Camilleri, 2023; Whittaker et al., 2023).

Future research can address the psychological and social dimensions of decision-making processes involving privacy with technological advances (Belanche et al., 2024; Beke et al., 2022; Liyanaarachchi et al., 2024; Wazid et al., 2024). Despite extensive research on the impact of deepfakes on consumers, their effect on businesses has been largely overlooked and warrants investigation (Mustak et al., 2023; Siegel et al., 2024).

This article provides a framework with a rich research agenda (see Table 3).

4.4. Conclusion

This study underscores the risk of deepfake technology to individuals, businesses, and society. Triangulating interview data, the study identifies critical themes such as data integrity and business risk, culminating in the formulation of “The business privacy calculus” conceptual model, providing a framework for navigating deepfake risks. The prosed model offers a pioneering perspective that extends beyond consumer-focused research. By focusing on the business context, we seek

Table 3
Future research agenda.

Research focus	Key topics
Business privacy calculus	• Test the efficacy of the business privacy calculus model in guiding organizations to make informed decisions that balance privacy risks and benefits. • Investigate the factors influencing organizational privacy calculus, such as industry norms, regulatory requirements, and competitive pressures. • Assess the compatibility of the business privacy calculus framework with evolving technological landscapes and anticipate future privacy risks and opportunities.
Consumer vulnerability	• Investigation of the psychological factors contributing to consumer vulnerability to deepfake manipulation. • Analysis of the effectiveness of educational interventions in reducing consumer susceptibility to deepfake scams. • Examination of demographic differences in susceptibility to deepfake influence and vulnerability.
Psychological reactance:	• Analysis of the emotional and cognitive responses elicited by exposure to deepfake content. • Exploration of strategies for effectively countering psychological reactance in deepfake awareness campaigns and interventions. • Evaluation of the long-term effects of repeated exposure to deepfake content on psychological well-being and behavior.
Industry & policy	• Exploration of the legal and ethical considerations surrounding businesses' responsibility to protect consumers from deepfakes. • Assessment of existing regulatory frameworks' adequacy in addressing deepfake-related privacy and security concerns. • Examination of the role of industry standards and self-regulatory initiatives in complementing government regulations in mitigating deepfake threats.
AI & deepfakes	• Analyze the utilization of sophisticated AI methods to identify fake content across various media types, enhancing accuracy through advanced analysis techniques. • Explore implementing continuous learning algorithms in AI systems, allowing adaptation to new deepfake methods. • Examine how collaborative platforms can facilitate cross-domain knowledge exchange between industries to build coordinated responses to deepfake threats.

to fill a significant void in existing literature and lay the groundwork for future inquiries in this emerging field.

CRediT authorship contribution statement

Giuseppe Vecchietti: Methodology, Formal analysis, Data curation. **Gajendra Liyanaarachchi:** Writing – original draft, Visualization, Resources, Conceptualization. **Giampaolo Viglia:** Writing – review & editing, Validation, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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